1.__________ characterizes the purity/impurity of an arbitrary collection of examples.

a) Fuzzy
b) Entropy Page 177
c) Destroy

2. Deductive learning working on --------- facts and knowledge

a) Fuzzy
b) Boolean
c) Existing Page 162
d) New

3. In unsupervised search a,_______represents some specific underlying pattern in the Data.

a) Cluster Page 190
b) Chunks
c) Fragments
d) Egment

4. ---------- is a form of unsupervised learning

a) Classificatin
b) Aggregation
c) Distribuation
d) Clustering Page 205

5. In planning phase, each state is represented in -------- logic

a) Boolean
b) Fuzzy
c) Predicate Page 197
d) Conceptual Page 197

6. In CLIPS, anything after a semicolon is a --------

a) Comment Page 137
b) Variable
c) Fact
d) Rule

7. The Linear model of software development has been successfully used in developing systems.

a) Expert Page 129
b) Software

c) Design
d) Logical

8. Reasoning in fuzzy logic is just a matter of generalizing the familiar _ logic.

a) Boolean Page 147
b) Complex
c) Coagnitive
d) Supervised

9. Which of the folllowing is correct command in CLIPS to see the list of Fact?

a) CLIPS> (show facts)
b) CLIPS> (view facts)
c) CLIPS> (list facts)
d) CLIPS> (facts)

10. In Candidate-Elimination algorithm version space is represented by two sets named:

a) G and S Page 173
b) G and F
c) S and F
d) H and S

11. While designing an expert system ,getting knowledge from the expert is called ------

a) Knowledge elicitation Page 130
b) Knowledge analysis
c) Inference networks
d) Planning

12. Which modifies the performance element so that it makes better decision?

a) Changing element
b) Learning element
c) Performance element
d) Observing element

13. Soft-computing is naturally applied in -------- learning applications

a) Machine Page 205
b) Algorithm
c) Knowledge
d) Information

14. A -------- is a person who posses the skill and knowledge to solve a specific problem in a manner superior to others.

a) Knowledge engineer
b) End-user
c) Naïve
d) Domain Expert Page 122

15. In backward chaining terminology, the hypothesis to prove is called the.

a) Proof
b) Goal Page 126
c) Plan
d) None of the given

16. Representation of the hypothesis space H for conjuction of attribute T (Temperature) and BP (Blood pressure) can be expressed

a) H = T , BP
b) H = T , T
c) H = BP , BP
d) H = <T , BP> Page 168

17. The hypothesis space we have two attributes T (temperture) and BP (Blood pressure) that can take 5 values each : L, N, H, ? and Ø. So there are -------- total hypotheses possible.

a) 10
b) 15
c) 20
d) 25 Page 170

18. The Entropy is 1 when the collection contains number of positive examples to/than negative examples.

a) Equal Page177
b) Greater
c) Less
d) Greater and Equal

19. Measure of the effectiveness of an attribute in classifying the training data is called

a) Information Gain Page 177
b) Measure Gain
c) Information Goal
d) Measure Goal

20. __________ logic lets us define more realistically the true functions that define real world scenarios.

a) Fuzzy Page 148
b) Classical
c) Boolean
d) Discrete

21. If the antecedent is only partially true, then the output fuzzy set is truncated according to the __________ method.

a) Intrinsic
b) Implication Page 153
c) Boolean
d) inclination

22. ______________ learning works on existing facts and knowledge and deduces new knowledge from the old.

a) Deductive Page 162
b) Inductive
c) Application
d) None of these

23.Outputs of learning are determined by the ____________

a) Application Page 161
b) Validation
c) Training
d) Algorithm

24. The tractable problems are further divided into structured and _____________ problems

a) Non-structured
b) Complex Page 166
c) Simple
d) difficult

25. A single layer perceptron cannot perform pattern classification on

a) linearly separable Page 186
b) linearly non-separable
c) intractabl
d) non-intractabl

26. In ANNs, Training is the heart of learning, in which finding the best that covers most of the examples is the objective.

a) Hypothesis Page189
b) Neuron
c) Agent
d) Operator

27. The primary function of soma and the enclosed nucleus in neuron is to ----------

a) Perform operation on incoming and outgoing data page 181
b) Keep neuron functional
c) Collect input data
d) Collect out data

In ANNs, MSE is known as

a) Most squared error
b) Mean squared error (Page 189)
c) Medium squared error
d) Mode squared error

28. A ---------- system can avoid any action that is just not possible at a particular state.

a) Planning Page 197
b) Fuzzy
c) Searching
d) Trained

29. --------- deals with procedures that extract useful information from static pictures and sequence of images.

a) Computer vision Page 203
b) Neural networks
c) Predicate logic
d) Descision support system

30. Reserch in Computer vision is conducted all mentioned areas EXCEPT

a) Object extraction
b) Automated navigation
c) Medical imaging
d) Digital Signal Proccessing Conversion

31. which one does not belong to the clustering algorithms?

a) Self-organizing maps(SOM)
b) K-means
c) Linear vector quantization
d) Volume based data analysis

32. An expert system may ------ the expert or assist the expert.

a) Ignore
b) Curew
c) Overload
d) Replace Page 113

33. The Entropy is --------- when the collection contains equal number of positive examples and negative examples.

a) 0
b) 1 page 177

34. A drawback of FIND-S is that, it assumes the consistency within the ------- set.

a) Training Page 173
b) Validation
c) Real
d) Application

35. --------- chaining, on the other hand is like an exhaustive search.

a) Backword
b) Parallel
c) Serial
d) Forward Page 128

36. Is the process by which the fuzzy sets that represent the outputs of each rule are combined into a single fuzzy set.

a) Aggregation Page157
b) Fuzzification
c) Implication
d) Defuzzification

37. Which of the combinations is possible to solve real world problems?

a) Genetic fuzzy
b) Neuro-fuzzy system
c) Packet and Frame

d) Genetic fuzzy and Neuro-fuzzy system Page 205

38. __________ is a prerequisite for any mature program of artificial intelligence.

a) Common Sense
b) Deep Learning
c) Machine learning Page 160
d) General Intelligence

39. ------- finds the maximally specific hypothesis possible within the version space

FIND-S Page 172

40. In all calculations involving entropy we define 0log 0 -------

0 Page 177

41. In Linear Model, a linear sequence of steps is applied repeatedly in an -------- fashion to develop the Expert System.

Iterative Page 129

42. In order debug the program in CLIPSwhich of following command is used?

Watch Page 135

43. In CLIPS, command is used to remove facts.

a) Facts
b) Retract Page # 134
c) Erase
d) Delete